

Development and Performance Analysis of The Second Generation 406 MHz EPIRB

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Abstract— COSPAS-SARSAT system which consists of space segment and ground segment provides a distress alert and the position information for a distressed ship or airplane with beacon equipment. Currently, COSPAS-SARSAT system provides the LEOSAR service which is called to the first generation, and the GEOSAR service as the second generation. The GEOSAR service provides position accuracy better than the LEOSAR service because the second generation 406 MHz beacon allows for the encoding of position data in the transmitted 406 MHz message, thus providing for quasi-real time alerts and position information through the GEOSAR service. ETRI developed Emergency Position Indicating Radio Beacon(EPIRB) as the second generation 406MHz beacon for use in maritime applications. Design and implementation aspects of the EPIRB developed by ETRI are presented. Also, performance is presented through the experiment.

Keywords—COSPAS-SARSAT, EPIRB, GEOSAR, 406MHz, Beacon

I. INTRODUCTION

The detection and location of an aircraft crash or maritime distress is of paramount importance to the search and rescue(SAR) teams and to the potential survivors. Studies show that while the initial survivors of an aircraft crash have less than a 10% chance of survival if rescue is delayed beyond two days, the survival rate is over 60% if the rescue can be accomplished within eight hours. Similar urgency applies in maritime distress situations, particularly where injuries have occurred. Furthermore, accurate location of the distress can significantly reduce both SAR costs and the exposure of rescue forces to hazardous conditions, and clearly improve efficiency. In view of this, Canada, France, Russia and the USA established the COSPAS-SARSAT satellite system to reduce the time required to detect and locate SAR events world-wide. The COSPAS-SARSAT system provides distress alert and location information to search and rescue(SAR) services anywhere in the world for aviators, mariners and land users in distress[1][2]. The COSPAS-SARSAT Low-altitude Earth Orbit (LEO) satellite system for search and rescue (LEOSAR) has been in operation since 1982. The system was complemented in 1998 with geostationary satellites (GEOSAR). GEOSAR service provides the enhancements to the COSPAS-SARSAT polar-orbiting system in terms of

alerting time advantage and the benefits to SAR services of this rapid alerting capability. Specially, the second generation 406 MHz beacon allows for the encoding of position data in the transmitted 406 MHz message, thus providing for quasi-real time alerts and position information from GEOSAR system[3].

II. SPECIFICATION AND DESIGN

A. Specification

Specification of the second generation 406MHz SAR beacon is defined in COSPAS-SARSAT technical document[4]. And, specification of EPIRB is defined in IMO document[5]. The second generation EPIRB is designed by above two specifications.

B. Design of EPIRB

Figure 1 shows the functional block diagram of second generation EPIRB.

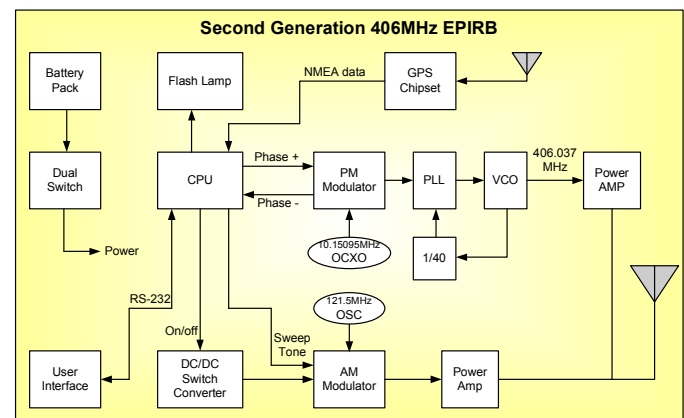


Figure 1. Functional Block Diagram of Second Generation SAR Beacon

The second generation 406MHz EPIRB receives NMEA data as its own position by GPS chipset. CPU generates a long message data(Bi-phase L, 144bit, 520ms) including the beacon's position data. PM modulator makes the message data phase modulate using 10.15095MHz OCXO. The message data is modulated with 406.037MHz by PLL circuit using 1/40 divider. And, it is transmitted to antenna through power

amplifier. After 406.037MHz beacon signal is transmitted, 121.5MHz homming signal is transmitted. The sweep tone signal is generated between 440Hz and 1100Hz. This signal is AM modulated by 121.5MHz OSC. AM modulator is operated by DC/DC switch converter controlled by CPU. The power is supplied by the battery pack through the dual switch. If the beacon signal is transmitted, the flash lamp is operated periodically. The beacon is configured and monitored by RS-232 protocol.

C. Housing Mechanism of EPIRB

Figure 2 shows the functional block diagram of the dual switch configuration of SAR beacon.

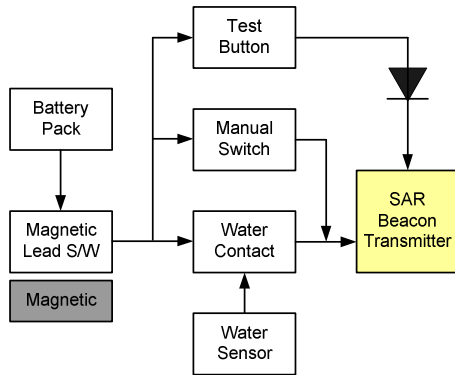


Figure 2. Functional Block Diagram of S Dual Switch Configuration

The second generation 406MHz SAR beacon is designed for marine. The dual switch configuration including the magnetic sensor and the water sensor is designed for preventing the beacon from transmitting wrong signal. The magnetic sensor on the housing is operated when the beacon is departed from the housing by water pressure. And the water sensor on the housing is operated when it is contact with the salt water such as sea. Manual switch and test button is for manual operation and test

III. IMPLEMENTATION

A. GPS Receiver

Figure 3 shows GPS receiver including the GPS chipset

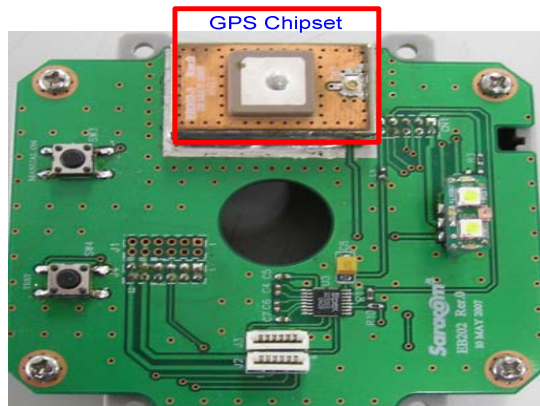


Figure 3. GPS Receiver

GPS receiver receives the GPS signal and calculates its own position. The position data is transformed into NMEA format, and then, GPS receiver transmits it to CPU.

B. Beacon Transmitter

Figure 4 shows the beacon transmitter including the 10.15095MHz OCXO.

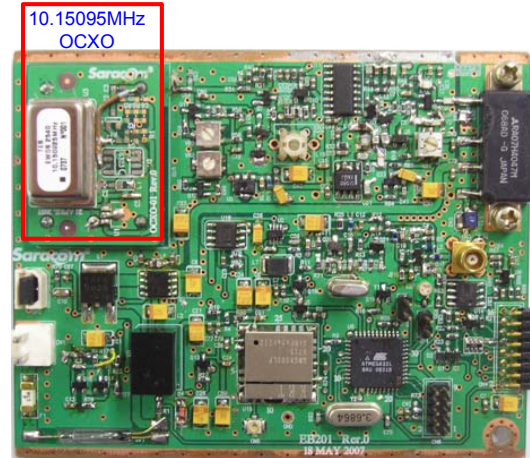


Figure 4. Beacon Transmitter

The beacon transmitter sends 406.037MHz beacon signal and 121.5MHz homming signal. Specially, 10.15095MHz OCXO which makes 406.037MHz by PLL is very important module of beacon because its clock stability and accuracy makes position accuracy in the LEOSAR system.

C. Layout of EPIRB

Figure 5 shows the layout of second generation EPIRB.

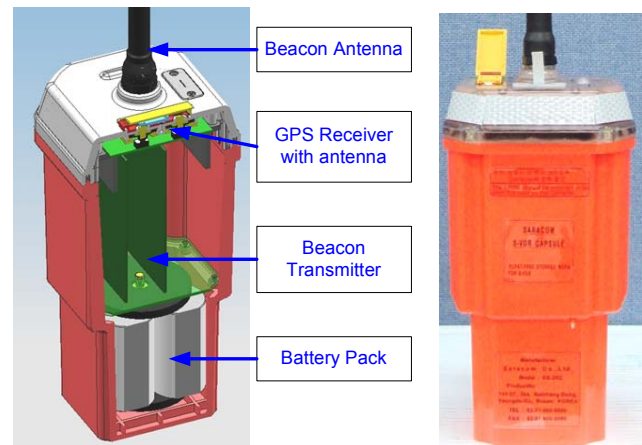


Figure 5. Second Generation SAR Beacon

The second generation EPIRB consists of the GPS receiver board, beacon transmitter board, the battery pack, and the antenna. The layout is designed for floating the beacon in the water and transmitting the beacon signal stably. So, the battery pack is disposed in the lower part and beacon transmitter board in the upper part. GPS receiver board is disposed in the top for

receiving the GPS signal. The right of figure 5 shows the prototype of second generation EPIRB.

D. Housing Mechanism of EPIRB

Figure 6 shows the housing mechanism of EPIRB.

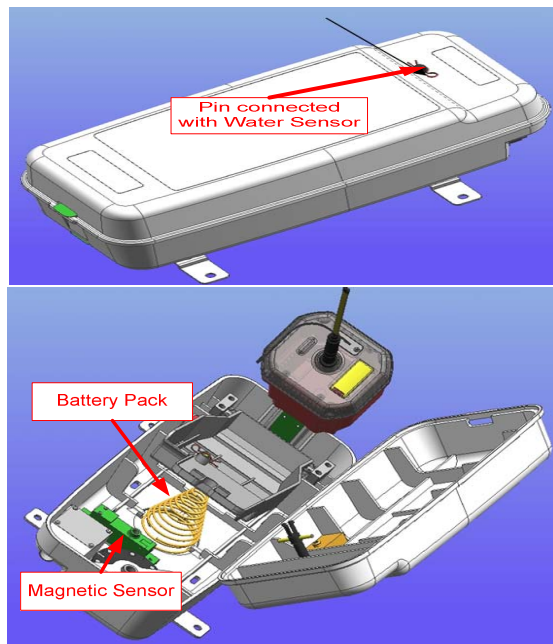


Figure 6. Housing Mechanism of EPIRB

The black pin is unlocked automatically when the water sensor on the housing is contact with the salt water. And then, the magnetic sensor on the housing is operated when the beacon is departed from the housing by water pressure. Finally, the spring on the housing send the beacon out.

IV. EXPERIMENT RESULTS

The test procedure of the second generation EPIRB complies with the COSPAS-SARSAT beacon type approval standard[6].

A. GPS Receiver

Figure 7 shows the experiment of GPS receiver.



Figure 7. Experiment of GPS Receiver

The experiment of GPS receiver is performed in the top of the building. The beacon antenna is grounded for preventing from transmitting the beacon signal during the test of GPS receiver. The NMEA data, that is, the position result of GPS receiver is stored in the memory of the beacon. Figure 8 shows the 4 results of NMEA data

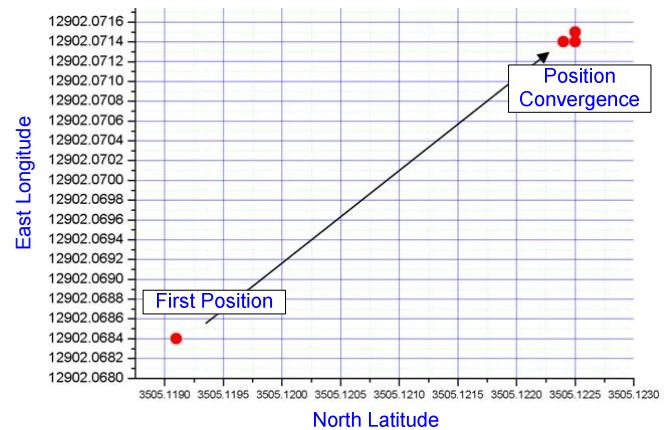


Figure 8. Result of NMEA data

All positions since the first position converge on the similar point, that is, east longitude is 129.020714° and north latitude is 35.051225° . So, the beacon message takes a valid position data since 3rd NMEA data for transmitting more accurate position.

Figure 9 shows the position plot in the Google map by using the above result.



Figure 9. Position Plot in Google Map

The position in the Google map pinpoints the similar point around the top of the building where this experiment was performed.

B. Beacon Transmitter

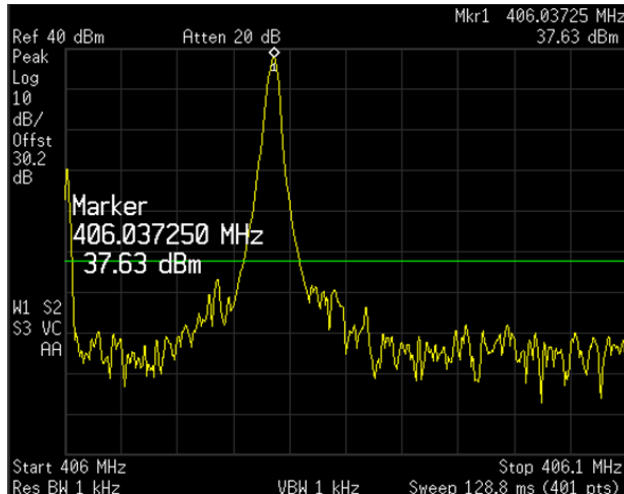


Figure 10. Experiment of 406.037MHz Beacon Signal

Figure 10 shows the experiment results of 406.037MHz beacon signal. The spectrum analyzer indicates the center frequency is 406.03725MHz and the transmitter power is 37.63dBm, that is, the specification value between 35dBm and 39dBm.

C. Housing Mechanism of EPIRB

Figure 10 shows the experiment of the housing mechanism.



Figure 11. Experiment of Housing Mechanism

The housing including EPIRB intentionally falls into the sea, being tied up with a heavy stone in order to make the environment like the water pressure and the salt water for the housing mechanism test.



Figure 12. EPIRB Operation

The housing is unlocked and the EPIRB come up to the surface of the sea. And the EPIRB is automatically operated.

V. CONCLUSION

COSPAS-SARSAT system provides the LEOSAR service which is called to the first generation, and the GEOSAR service as the second generation. The GEOSAR service provides position accuracy better than the LEOSAR service because the second generation 406 MHz beacon allows for the encoding of position data in the transmitted 406 MHz message, thus providing for quasi-real time alerts and position information through the GEOSAR service. ETRI developed Emergency Position Indicating Radio Beacon(EPIRB) as the second generation 406MHz beacon for use in maritime applications. This paper presented design and implementation aspects of the EPIRB. Also, performance was presented through the experiment. This EPIRB passed the authentication test in the test organization of the COSPAS-SARSAT and will be preparing for a sale as the commercial product.

ACKNOWLEDGMENT

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